





BUYSURE AI

An Intelligent Shopping Assistant for Trustworthy Product Evaluation

Personalized Purchase Decision Support: Buy / Wait / Avoid

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PROJECT GUIDE

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INTRODUCTION

- **Online shopping offers convenience and a huge choice, but picking the right product is difficult.**
- **Fake or misleading reviews, changing prices, unreliable sellers and many similar products add uncertainty.**
- **An average star rating alone can hide repeated problems with battery, durability, delivery or support.**
- **BUYSURE AI uses NLP and ML to analyze product details, reviews, ratings, prices, features and comparable products.**
- **It gives a clear, explainable recommendation (Buy, Wait or Avoid) based on the user's budget and requirements.**

ABSTRACT

- **BUYSURE AI is an AI-based shopping assistant for safer, smarter purchase decisions.**
- **Analyzes product details, reviews, ratings, prices, features and comparable products.**
- **NLP cleans review text, estimates sentiment and extracts positive and negative aspects.**
- **ML flags suspicious or low-credibility reviews using text, rating, metadata and behavior signals.**
- **Compares products against the user's budget, preferences and requirements.**
- **Gives a Buy / Wait / Avoid recommendation with key reasons, risks and trade-offs.**

BASE PAPER

- **Reference: “Fake Reviews Detection: A Survey,” IEEE Access, vol. 9, pp. 65771–65802, 2021**
- **Contribution: Surveys fake-review detection using linguistic, behavioral and statistical signals**
- **Methodology: Compares traditional ML and deep-learning approaches; transformer representations give strong baselines**
- **Dataset: Survey study; reviews existing work rather than proposing a new product dataset**
- **Limitation: Stops at review classification: no product comparison, personalization or purchase decision**
- **Our extension: Aspect extraction, budget-aware comparison and an explainable Buy / Wait / Avoid output**

LITERATURE SURVEY (1/2)

Author & Title	Methodology	Limitations	Our Contribution to Overcome the Limitations
[3] "Fake Reviews Detection Using Deep Learning: A Survey" (IMSA, 2024)	Surveys neural methods that learn contextual relationships in deceptive-review text.	Class imbalance, domain shift, generated text and adversarial behavior remain open challenges.	Treats flags as risk indicators and weights evidence by credibility instead of trusting one model.
[4] "Fake Review Detection in E-Commerce Using ML and NLP" (ICICI, 2025)	Sentiment and n-gram features plus reviewer behavior and time patterns; KNN, LR and SVM classifiers.	Focused on detection only; no aspect-level pros and cons or product comparison.	Feeds credibility into aspect summaries, comparison and the final recommendation.
[5] "A Hybridized Approach for Enhanced Fake Review Detection" (IEEE TCSS, 2024)	Attention-based recurrent model with imbalance-aware training.	Detection-centric; does not turn results into a user-facing purchase decision.	Keeps the review model replaceable inside a modular decision-support pipeline.
[6] M. Chelliah, S. Sarkar, "Product Recommendations Enhanced with Reviews" (ACM RecSys, 2017)	Uses sentiment analysis, topic modeling and embeddings to exploit review text in recommenders.	Assumes review text is trustworthy; no suspicious-review screening.	Screens review credibility first, then uses reviews for feature-level evidence.

LITERATURE SURVEY (2/2)

Author & Title	Methodology	Limitations	Our Contribution to Overcome the Limitations
[7] "Multi-Criteria Review-Based Recommender System: The State of the Art" (IEEE Access, 2019)	Argues one overall rating cannot capture fine-grained preferences; uses review-based criteria.	Does not address fake or low-credibility reviews or budget constraints.	Uses aspect-level criteria with hard budget and mandatory-requirement checks.
[8] "A Survey of Explainable E-Commerce Recommender Systems" (WI-IAT, 2022)	Reviews methods that give understandable reasons for recommendations.	Explanations are not tied to review credibility or evidence confidence.	Generates explanations from stored evidence with pros, cons, risks and confidence.
[9] "A Review of Text-Based Recommendation Systems" (IEEE Access, 2021)	Reviews recommenders that use textual content such as descriptions and reviews.	General recommendation focus; no Buy / Wait / Avoid style decision support.	Converts text evidence into a clear, personalized purchase verdict.
[10] "Sentiment Analysis-Based Product Review System for Enhanced Recommendations" (AIDE, 2025)	Sentiment analysis of product reviews to improve recommendations.	Sentiment alone misses feature context and review reliability.	Adds aspect extraction and credibility weighting to sentiment.

OBJECTIVES

- **Collect and organize product details, prices, ratings, features and seller data.**
- **Preprocess customer reviews and extract sentiment using NLP.**
- **Detect suspicious review patterns as risk indicators, not accusations.**
- **Extract positive and negative product aspects from genuine feedback.**
- **Compare products using normalized features and user priorities.**
- **Personalize results using budget and mandatory requirements.**
- **Produce an interpretable Buy / Wait / Avoid decision with reasons.**
- **Keep the design modular for image analysis, price tracking and seller-risk checks.**

BUYSURE AI — SYSTEM ARCHITECTURE

BuySure AI Platform

1. User & Input Layer

User

Budget • Mandatory needs • Feature priorities

2. Data Acquisition Layer

- Product APIs / Catalogues
- Prices & Specifications
- Seller Information
- Customer Reviews

3. Processing & Storage Layer

Preprocessing

- Text cleaning
- Deduplication
- Unit normalization

Product & Review Store

Processed Features

4. AI Intelligence Layer

Review Intelligence

- Sentiment
- Aspect extraction
- Credibility

Product Matching

- Requirements
- Feature alignment
- Price fit

Evidence Fusion

- Weighted evidence
- Risk penalty
- Confidence

5. Decision & Output Layer

Decision Engine

- Hard constraints first
- Buy / Wait / Avoid rules

Report & Dashboard

- Pros / Cons
- Risks
- Confidence
- Alternatives

User Result

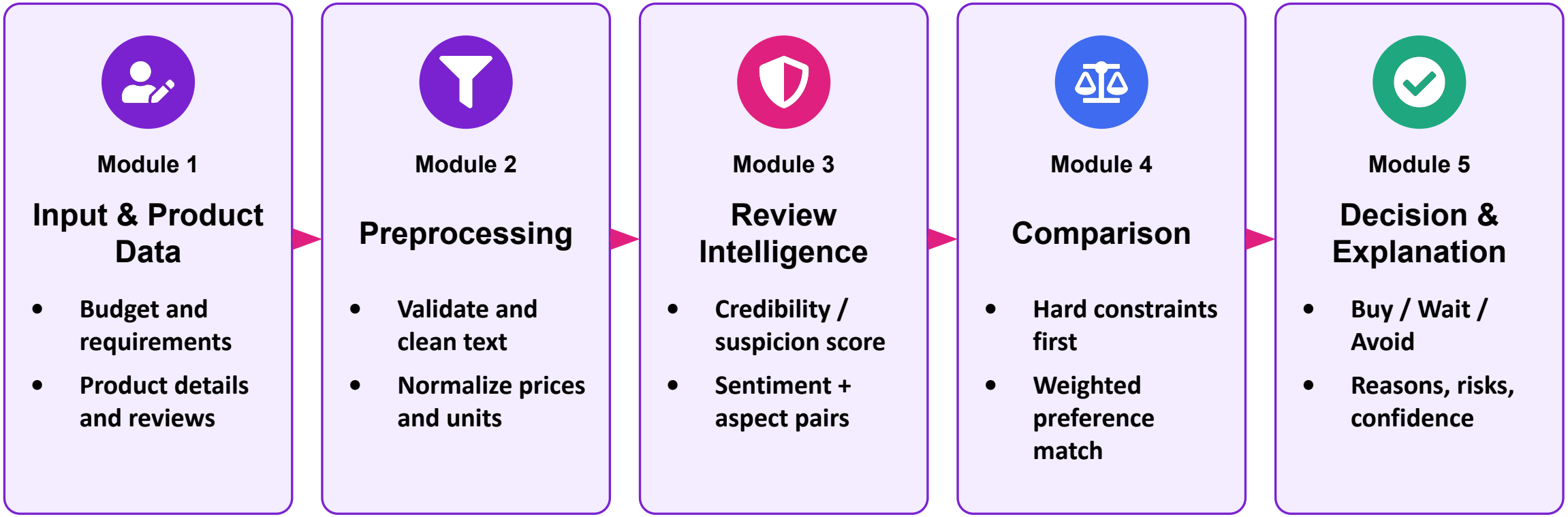
- Buy
- Wait
- Avoid

Cross-Cutting Services

Validation & Error Handling | Logging | Security | Model Evaluation

Future Extensions: Image / Screenshot Analysis • Price Tracking • Seller-Risk Detection • Better Alternative Discovery

MODULE FLOW



Raw evidence is kept separate from derived signals, so every recommendation can be traced back to the reviews and data that caused it.

BUY / WAIT / AVOID DECISION



BUY

Mandatory requirements are met and no dominant risk or credibility concern exists.



WAIT

Product may suit the user, but price uncertainty, thin evidence or an unresolved trade-off weakens the decision.



AVOID

A major requirement fails, or review, seller or quality evidence shows a substantial risk.

Evidence-fusion score

$$\text{Score}(p) = w1 \cdot M(p) + w2 \cdot R(p) + w3 \cdot V(p) + w4 \cdot Q(p) - w5 \cdot K(p)$$

M = requirement match, R = review evidence, V = value for money, Q = quality evidence, K = risk penalty. Weights are configurable and thresholds are tuned during evaluation.

RESULT & EVALUATION PLAN

BUYSURE AI Report (illustrative layout)

DECISION: BUY Evidence confidence: example

Budget fit	Within stated budget
Requirements	Mandatory needs satisfied
Review health	Suspicious-review share shown
Pros	Strong priority-feature feedback
Cons	Repeated concerns listed
Action	Compare, wait for price or proceed

Evaluation plan

- Review credibility: precision, recall, F1, confusion matrix
- Sentiment and aspects: class-wise F1
- Normalization: field-level correctness
- Recommendation: scenario tests vs expert labels
- Explanation: human ratings of clarity
- Robustness and latency checks

The sample report is a layout mock-up. Measured accuracy results will be added after the dataset and implementation are finalized.

FUTURE SCOPE

- **Image analysis: extract visible specifications from product images and screenshots**
- **Price tracking: timestamped prices and target-price alerts**
- **Seller risk detection: use permitted seller ratings and policy signals**
- **Better alternatives: find products that satisfy unmet requirements**
- **Multilingual reviews: support regional and code-mixed review text**
- **Feedback loop: user feedback with privacy safeguards**

CONCLUSION

- **BUYSURE AI turns scattered product data and reviews into one understandable decision report.**
- **Review credibility, sentiment, aspects, budget and requirements are combined in a transparent pipeline.**
- **Separating raw evidence from the final decision keeps every recommendation traceable.**
- **Next step: connect the architecture to a documented dataset, implement it and report measured results.**

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[1] S. R. Srikowsalya, S. Sujith and S. Lingesh, "BUYSURE AI: Project Abstract," Dept. of CSE, VISTAS, 2026.

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